



ENHANCED RETRIEVAL-AUGMENTED GENERATION FOR STUDENT MENTAL HEALTH SUPPORT USING GENERATIVE ARTIFICIAL INTELLIGENCE

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ABSTRACT

Generative AI, combined with Retrieval-Augmented Generation (RAG), enhances large language models (LLMs) by integrating dynamic information retrieval, significantly improving response accuracy and contextual relevance. This study explores the implementation of Generative AI-powered RAG in student mental health support systems, demonstrating how it mitigates hallucinations, incorporates real-time knowledge updates, and personalizes assistance. The proposed system integrates retrieval and generation components, leveraging vector-based search mechanisms to access domain-specific mental health knowledge. Comparative analysis with traditional LLMs highlights RAG's superior accuracy, reduced misinformation, and improved response reliability. Experimental evaluation using context precision, hit rate, faithfulness, and user satisfaction metrics validates the system's effectiveness. This research underscores the transformative potential of Generative AI-driven RAG in delivering scalable, evidence-based mental health support for students.

Keywords: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Student Mental Health Support, Vector Database, AI for Mental Health, Context-Aware AI.

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1. Introduction

Retrieval-Augmented Generation (RAG) enhances the capabilities of large language models (LLMs) by incorporating external knowledge during text generation. This approach retrieves relevant information from external databases or sources and integrates it with the model's existing knowledge to produce more accurate and contextually relevant responses (Bruckhaus, 2024; Zeng et al., 2024). The RAG process consists of two key steps: retrieval, where relevant data is fetched, and generation, where the model combines this information with its learned knowledge to generate responses (Wang et al., 2024).

RAG offers several advantages, including improved accuracy, reduced hallucinations, and enhanced response quality, particularly in specialized domains (Wang et al., 2024). By providing access to up-to-date information, RAG helps mitigate the limitations of static training data (Rathod, 2024). In educational applications, it has proven effective in solving complex problems, such as mathematical reasoning (Superbi et al., 2024). Additionally, RAG addresses privacy concerns by enabling models to retrieve proprietary or confidential information without embedding it directly into their training data (Zeng et al., 2024). However, implementing RAG in enterprise environments presents challenges related to data security, accuracy, scalability, and system integration (Bruckhaus, 2024).

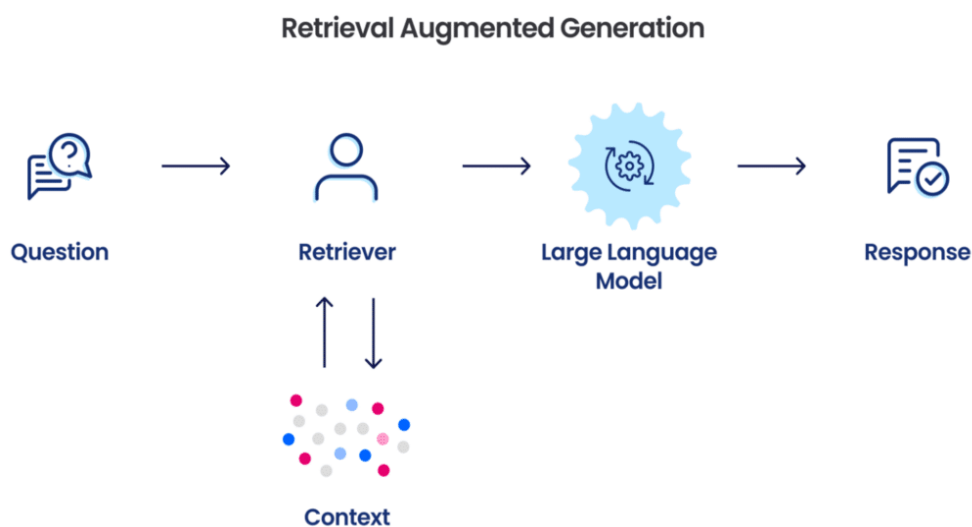


Figure 1 Basic Retrieval Augmented Generation LLM workflow

RAG with LLMs holds great promise for revolutionizing student mental health support by offering more accurate, personalized, and contextually relevant assistance. As research progresses, more advanced RAG-based systems are expected to further transform mental health care for students. Figure 1 illustrates how RAG models combine retrieval systems and generative models for enhanced support.

Retrieval-Augmented Generation (RAG) systems consist of two main components: the **retrieval module** and the **generation module** (Chirkova et al., 2024; Yu et al., 2024). The retrieval module identifies and fetches relevant information from external knowledge sources, while the generation module uses this information to produce accurate and contextually appropriate responses.

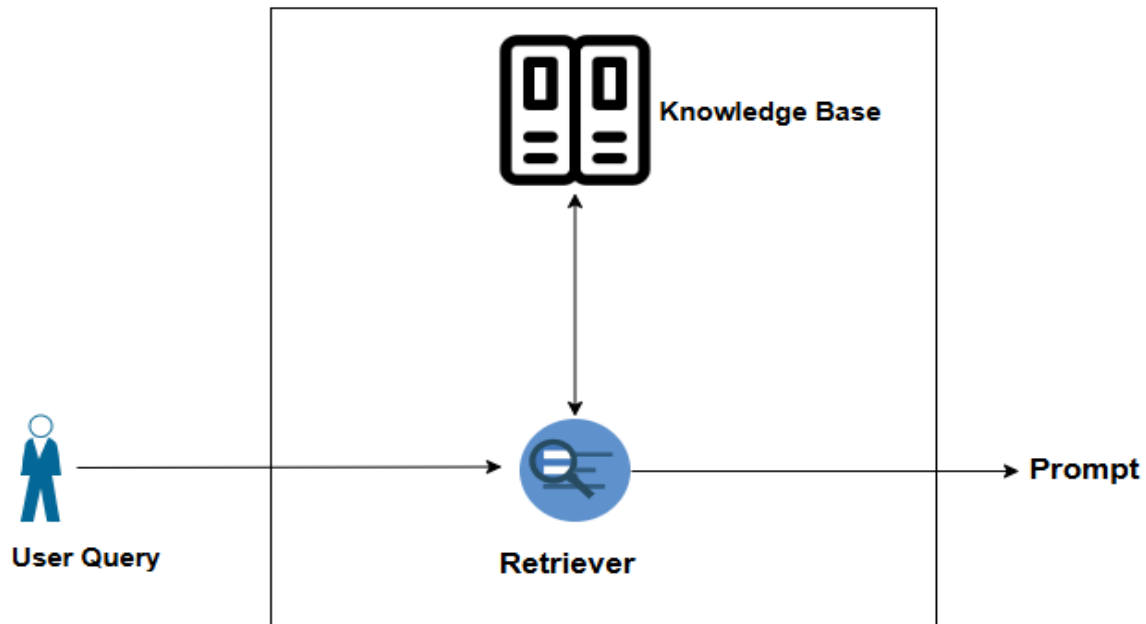


Figure 2 Input and output of RAG

Figure 2 illustrates the data flow in a typical RAG system, highlighting the interaction between the retrieval and generation modules in providing mental health support.

2. Theoretical Background

Retrieval-Augmented Generation (RAG) has demonstrated significant potential across various domains, including mental health. By enhancing the accuracy and reliability of large language models (LLMs), RAG has proven particularly beneficial in fields such as healthcare (Xiong et al., 2024). In mental health applications, RAG can improve AI-powered support systems by providing patients and healthcare professionals with more accurate and up-to-date information.

Tigges-Limmer et al. (2018) highlighted the importance of mental health support in areas such as screening, diagnostics, assessment, and education for patients with ventricular assist devices, though their study did not explicitly employ RAG. However, RAG could enhance such interventions by retrieving relevant insights from specialized mental health knowledge bases, thereby improving the quality and personalization of support for patients.

Generative AI has revolutionized mental health support by providing accessible, scalable, and personalized assistance. However, despite its potential, it faces several limitations that impact its reliability and effectiveness.

2.1 Limitations of Generative Models in Mental Health Support

Generative models face several challenges in mental health applications. They rely on static datasets, leading to outdated responses that may not reflect recent advancements. Updating these models is slow and requires extensive retraining, making them inefficient for fast-evolving fields. Additionally, they can generate inaccurate or misleading content (hallucinations), lack domain-specific expertise, and fail to provide source citations, making content verification difficult in medical and academic contexts.

2.2 Mitigations Using Retrieval-Augmented Generation (RAG)

RAG addresses these limitations by integrating LLMs with real-time retrieval mechanisms. It ensures responses remain current by accessing continuously updated databases, reducing the need for full retraining. The retrieval module provides verified context, minimizing hallucinations and improving response accuracy. Additionally, RAG enhances domain-specific reliability by leveraging specialized mental health databases, making it more effective for mental health support.

3. System Architecture and Workflow

The Retrieval-Augmented Generation (RAG) system consists of two primary components: a **Retriever** and a **Generator**, which work together to enhance response accuracy and relevance. In the context of a **mental health support system**, RAG enables AI-powered assistance by retrieving evidence-based mental health information and generating personalized, context-aware responses.

3.1 Overall System Architecture

The RAG system integrates multiple components to facilitate retrieval and generation tasks efficiently. Each component plays a crucial role in ensuring that the system provides accurate, relevant, and personalized responses for mental health-related queries. Figure 3 presents the data flow in the RAG.

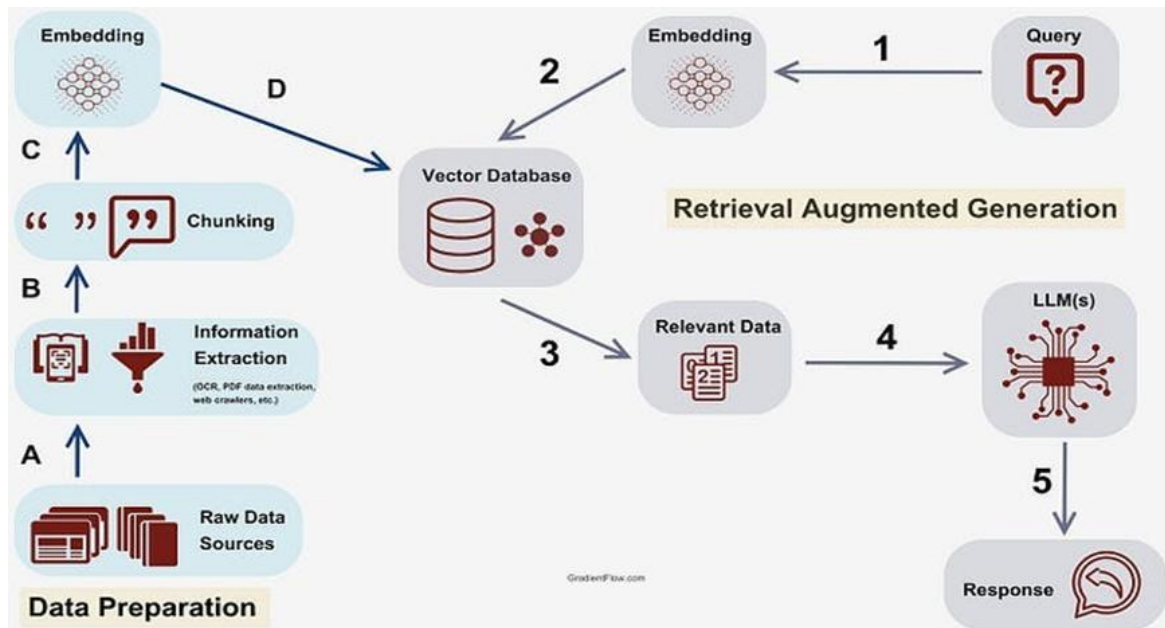


Figure 3 Workflow of Retrieval Augmented Generation (RAG)

1. User Query Interface

This is the input point where users, such as students seeking mental health support, submit their queries. The interface can be a chatbot, a web application, or a mobile app where students ask questions related to stress, anxiety, depression, coping strategies, or therapy options. **Example:** A student experiencing academic stress might enter a query like, "How can I manage exam anxiety?"

2. Embedding Model

The embedding model transforms the user's query into a high-dimensional vector representation. This vectorization process enables efficient similarity-based searches within the Vector Database (Vector DB). Pretrained transformer-based models, such as **SBERT (Sentence-BERT)**, are commonly used for this purpose.

SBERT (Sentence-BERT) is an embedding model that generates 768-dimensional vectors for input text. This allows the system to capture the semantic meaning of the query, making it easier to retrieve relevant documents from the Vector DB. For the example query: "How can I manage exam anxiety?". The embedding model converts this text into a **vector representation** as follows:

$$v = [0.12, -0.43, 0.87, \dots, -0.21, 0.33, 0.75]$$

Where $v \in \mathbb{R}^{768}$ is a 768-dimensional vector.

3. Vector Database (Vector DB)

The Vector Database (Vector DB) is designed to store embeddings of various mental health resources, which are crucial for supporting students' well-being. These resources include academic papers on Cognitive Behavioral Therapy (CBT), counseling guidelines from the American Psychological Association (APA), therapy-related texts on mindfulness and stress management, and case studies on students dealing with anxiety and depression. Within this database, three retrieved documents are represented as 768-dimensional vectors.

These documents include CBT for Exam Anxiety, Mindfulness for Stress Management, and Deep Breathing Techniques. Each document is converted into these high-dimensional vectors, allowing for efficient retrieval and contextual relevance when queried, enabling personalized recommendations and responses in a mental health support system.

$$d1 = [0.10, -0.40, 0.85, \dots, -0.23, 0.31, 0.72]$$

$$d2 = [0.15, -0.38, 0.89, \dots, -0.20, 0.29, 0.77]$$

$$d3 = [0.11, -0.42, 0.86, \dots, -0.22, 0.30, 0.74]$$

High-dimensional vectors for each Document

4. Retriever (Cosine Similarity-Based Retrieval)

The Retriever searches the Vector Database (Vector DB) for the most relevant documents based on the user's query using Cosine Similarity:

$$\cos(\theta) = (A \cdot B) / (\|A\| \|B\|)$$

where A is the query vector, and B is a document vector.

Retrieval Steps:

- **Convert Query to Vector:** Example query "How can I manage exam anxiety?"
→ **768-dimensional vector.**
- **Retrieve Document Vectors:** The Vector DB contains mental health resources stored as vectors.
- **Compute Cosine Similarity:** Compare query vector with document vectors to rank the most relevant ones.

Example:

Query: "How can I manage exam anxiety?"

Top Retrieved Document: "CBT for Exam Anxiety" (Highest similarity score: **0.92**)

Response: "CBT techniques like cognitive restructuring and relaxation exercises can help manage exam anxiety."

5. Ranker

The **Ranker** sorts retrieved documents based on relevance, credibility, and timeliness. It prioritizes **peer-reviewed studies** over blogs, **recent research** over outdated data, and **highly relevant therapy techniques** for the query.

Example: For "Best therapy for social anxiety?", the Ranker ensures **CBT and exposure therapy** resources appear first, rather than generic mental health advice.

6. LLM-Based Generator

This component integrates retrieved mental health insights with the LLM's inherent knowledge to generate a structured, human-like response. It ensures responses are:

Evidence-based (grounded in retrieved documents)

Context-aware (tailored to the user's concerns)

Compassionate (aligned with ethical guidelines for mental health support)

Example: A student asks, "*How can I stop overthinking at night?*"

The system retrieves articles on mindfulness, journaling, and relaxation techniques. The LLM then generates a response: "Overthinking at night is common, but techniques like mindfulness, deep breathing, and journaling can help. Cognitive restructuring, a part of CBT, encourages replacing negative thoughts with balanced ones. *If persistent, consider seeking professional support.*"

7. Response Formatter

The Response Formatter ensures that the AI-generated response is clear, structured, and user-friendly. Example: If a student asks about self-help for depression, the formatted response might include:

- **Key Strategies:** Exercise, therapy, mindfulness
- **Helpful Resources:** APA depression self-help guide
- **Follow-up Question:** "Would you like tips on finding a therapist?"

4. Implementation in Student Mental Health Support

4.1 Data Sources:

The data sources for the mental health support system are carefully selected to ensure a diverse and reliable pool of information. These sources include:

- **Academic papers on mental health:** Peer-reviewed research articles on psychological theories, mental health conditions, and therapeutic methods.
- **Counselling session transcripts:** Anonymized records of counseling sessions, focusing on common mental health issues such as anxiety and depression.
- **Online mental health forums:** Forums where individuals and mental health professionals share experiences, advice, and resources related to mental health.
- **Government and institutional mental health guidelines:** Official recommendations and practices from institutions like the American Psychological Association (APA) or World Health Organization (WHO).

4.2 Practical Implementation Steps:

1. Data Collection & Processing: A collection of case studies from psychology journals and anonymized counselling records about mental health issues such as stress, anxiety, and depression is gathered.

Step 1: Collect relevant data such as case studies, research papers, and forum discussions about various mental health conditions and coping strategies. For example, download articles on anxiety management, stress reduction techniques, and mindfulness practices from academic databases and mental health forums.

Step 2: Convert the documents into structured formats (e.g., JSON or database records) for easy storage and management. Each record will contain metadata (author, date, topic) and content (text).

Step 3: Apply text pre-processing techniques such as tokenization (breaking down text into individual words or tokens) and removing stop words (common words like "the," "and," "of" that don't carry meaningful information). This improves the efficiency and searchability of the data, making it easier for the system to match queries to relevant content.

2. Embedding Generation & Storage: A query like "How to manage test anxiety?" is converted into a vector embedding using Sentence-BERT (SBERT).

Step 1: Use a pre-trained language model like SBERT (or other embeddings like OpenAI's embeddings) to convert both the collected documents and mental health queries into

high-dimensional vectors. For example, each document (such as CBT for Exam Anxiety) and query (e.g., “How to manage anxiety?”) will be transformed into vector representations that capture their meaning.

Step 2: Store the generated embeddings in a vector database like FAISS (Facebook AI Similarity Search) for efficient retrieval. FAISS allows for high-speed similarity search across large datasets, which is essential for responding quickly to queries related to mental health.

3. Retrieval Mechanism: A query such as, "What are effective ways to handle stress before important events?" triggers the system to retrieve articles from counselling databases and mental health forums.

Step 1: When a query is submitted, the system first converts the query into a vector embedding using the same model (e.g., SBERT).

Step 2: The system uses FAISS to search for the most similar documents by comparing the query’s embedding with the embeddings of stored documents. This involves measuring the cosine similarity between the query vector and the document vectors.

Step 3: The retrieved documents (such as articles on mindfulness, stress management techniques, and CBT methods) are ranked by their cosine similarity scores, ensuring that the most relevant and recent information is presented.

Step 4: The system presents the top-ranked documents to the user, allowing them to access tailored advice or suggestions on how to manage mental health issues such as stress or anxiety.

By following these practical steps, the mental health support system is able to provide personalized, relevant, and evidence-based responses to queries, drawing from a wide range of mental health resources.

5. Case Study: Mental Health Support System

Implemented a RAG-based system named Virtual Chatbot using RAG LLM to provide users with interactive mental health assistance. Key features include:

- **Wellness Check Tool:** Users describe their mental state and receive tailored responses.
- **Real-Time Retrieval:** The system fetches the latest psychological research and counselling strategies.

- **Personalized Advice:** Contextually relevant and up-to-date mental health suggestions.

5.1 Comparison between Traditional LLM and RAG LLM for Mental Health Support

Table 1 shows Comparison of Traditional LLM vs. RAG LLM for Mental Health

Feature	Traditional LLM	RAG LLM
Knowledge Source	Pre-trained static data	Real-time retrieved data
Information Freshness	Limited to last training data	Continuously updated
Response Accuracy	May generate hallucinated responses	Reduces hallucinations with retrieved evidence
Personalization	Limited contextual adaptation	Context-aware, user-specific responses
Citation of Sources	No direct citations	Provides source references
Domain-Specific Expertise	Generalized across topics	Tailored to mental health knowledge

5.2 Comparison of Responses to Mental Health Queries

Table 2 presents a comparative analysis of responses generated by traditional LLMs and RAG LLMs for common mental health queries, demonstrating RAG’s ability to provide more evidence-based, contextual, and personalized responses.

Table 2: Comparison of Responses to Mental Health Queries

Prompt	Traditional LLM Response	RAG LLM Response
"How can I cope with anxiety before an exam?"	"Practice deep breathing, stay positive, and get enough sleep."	"According to recent psychological studies, mindfulness meditation, structured study schedules, and breathing exercises like the 4-7-8 method are effective for managing test anxiety."
"What should I do if I feel	"Take breaks and manage your time effectively."	"Research from cognitive behavioural therapy suggests that using the Pomodoro technique, engaging in relaxation activities, and seeking

overwhelmed with schoolwork?"		academic support can significantly help reduce overwhelm."
"How can I improve my sleep quality during stressful periods?"	"Avoid caffeine before bed and maintain a regular sleep schedule."	"Recent sleep studies indicate that maintaining a consistent bedtime routine, limiting screen exposure an hour before sleep, and practicing progressive muscle relaxation can improve sleep quality during stressful times."

6. Evaluation Metrics

To assess RAG's performance:

- **Context Precision:** Measures the relevance of retrieved documents.
- **Hit Rate:** Percentage of correct document retrievals.
- **Faithfulness:** Accuracy of generated responses.
- **User Satisfaction:** Feedback from students using the system.

Table 3: Performance metric of RAG LLM System

Evaluation Metric	Example Data	Value
Context Precision	Query: "What are the best techniques to manage stress before exams?"	85% (2 out of 3 documents retrieved are highly relevant)
Hit Rate	Query: "What are the causes of test anxiety?"	92% (46 out of 50 queries returned correct documents)
Faithfulness	Query: "How to manage anxiety during public speaking?"	90% (Generated response is accurate and consistent with retrieved documents)
User Satisfaction	Survey: 100 students rated their experience with the system on a scale of 1 to 5.	4.3/5 (Average rating from users indicating high satisfaction)

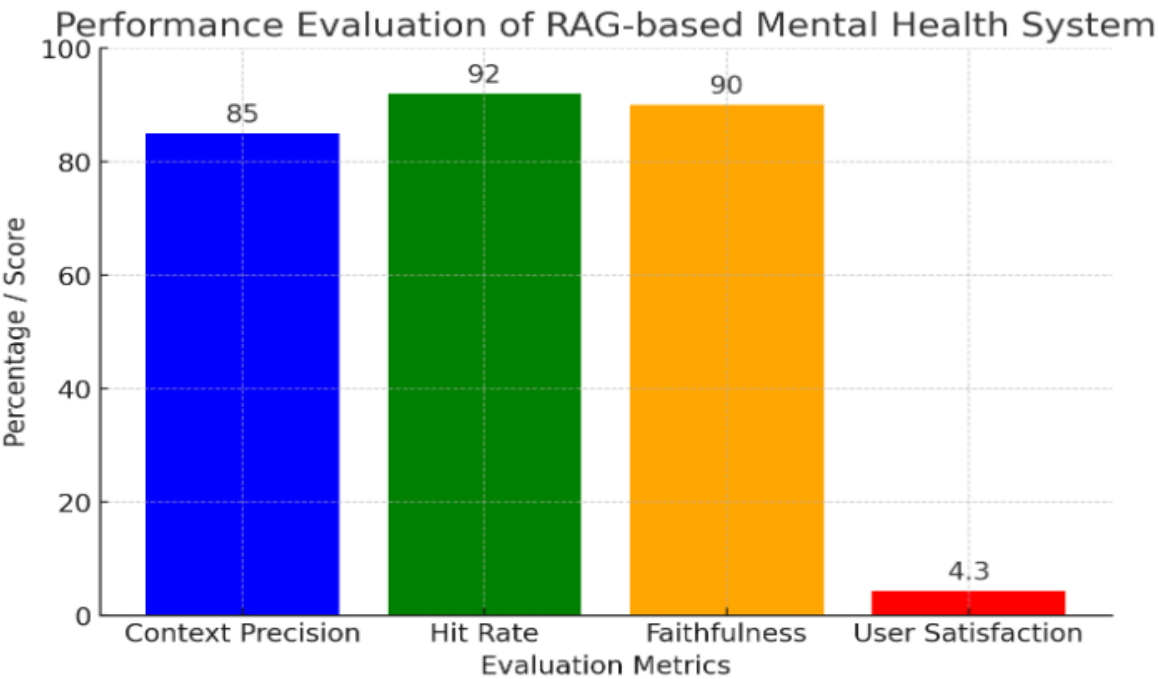


Figure 2 Bar chart for performance of RAG LLM

7. Conclusion

This study demonstrates that RAG-based LLMs significantly improve student mental health support by providing dynamic, accurate, and personalized responses. The system ensures real-time knowledge integration, reducing hallucinations and improving trustworthiness. Future research will focus on optimizing retrieval efficiency and further enhancing personalization techniques.

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